Object Oriented Programming (CSN13250)
ASSIGNMENT-1

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Q1. You are required to maintain your own record that includes your name, father's name, date of birth, age (computed with respect to the current date). The age includes years, months and days. You are required to show your record on the display screen such as: for displaying your name, it should display like "Name of the student:" followed by its value. Display of each entity should starts with new line. Use scanner class for getting the input from keyboard.

A1. CODE:

```
import java.time.LocalDate;
import java.util.Scanner;

public class P1 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);

        System.out.print("Enter your Name: ");
        String name=sc.next();
        System.out.print("Enter your Father's Name: ");
        String fatherName=sc.next();


        System.out.print("Enter DOB (yyyy-mm-dd): ");
        String dob=sc.next();
        String[] parts=dob.split("-");
        int byear=Integer.parseInt(parts[0]);
        int bmonth=Integer.parseInt(parts[1]);
        int bday=Integer.parseInt(parts[2]);

        LocalDate today=LocalDate.now();

        int cyear=today.getYear();
        int cmonth=today.getMonthValue();
        int cday=today.getDayOfMonth();

        int age=cyear-byear;

        if(cmonth<bmonth || (cmonth==bmonth && cday<bday)){age--;}
        System.out.println("Name = "+name);
        System.out.println("Father's Name = "+fatherName);
        System.out.println("DOB = "+dob);
        System.out.println("Age = "+age+" years");
    }
}
```

OUTPUT:

```
Enter your Name: ABCD
Enter your Father's Name: XYZ
Enter DOB (yyyy-mm-dd): 2007-09-15
Name = ABCD
Father's Name = XYZ
DOB = 2007-09-15
Age = 18 years
```

Q2. You are required to repeat the problem 1 with constraint that, use buffer class (bufferedReader) in place of Scanner class for taking the input from keyboard.

A2. CODE:

```
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.IOException;
import java.time.LocalDate;

public class P2 {
   public static void main(String[] args) throws IOException {
      BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

      System.out.print("Enter your Name: ");
      String name = br.readLine();
      System.out.print("Enter your Father's Name: ");
      String fatherName = br.readLine();

      System.out.print("Enter DOB (yyyy-mm-dd): ");
      String dob = br.readLine();
      String[] parts = dob.split("-");
      int byear = Integer.parseInt(parts[0]);
      int bmonth = Integer.parseInt(parts[1]);
      int bday = Integer.parseInt(parts[2]);

      LocalDate today=LocalDate.now();

      int cyear=today.getYear();
      int cmonth=today.getMonthValue();
      int cday=today.getDayOfMonth();

      int age=cyear-byear;

      if(cmonth<bmonth || (cmonth==bmonth && cday<bday)){age--;}
      System.out.println("Name = "+name);
      System.out.println("Father's Name = "+fatherName);
      System.out.println("DOB = "+dob);
      System.out.println("Age = "+age+" years");
   }
}
```

OUTPUT:

```
Enter your Name: ABCD
Enter your Father's Name: XYZ
Enter DOB (yyyy-mm-dd): 2007-09-15
Name = ABCD
Father's Name = XYZ
DOB = 2007-09-15
Age = 18 years
```

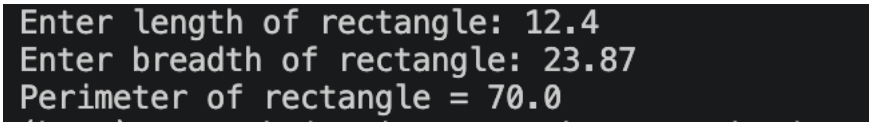
Q3. You are required to compute perimeter of a rectangle where length and breadth are taken as float and double respectively from the keyboard. Consider length and breadth as integer while computing its perimeter and computed perimeter is displayed as double.

A3. CODE:

```
import java.util.Scanner;

public class P3{
   public static void main(String[] args) {
      Scanner sc=new Scanner(System.in);
      float length;
      double breadth;
      System.out.print("Enter length of rectangle: ");
      length=sc.nextFloat();
      System.out.print("Enter breadth of rectangle: ");
      breadth=sc.nextDouble();
      int l=(int)length;
      int b=(int)breadth;
      double perimeter=2*(l+b);
      System.out.println("Perimeter of rectangle = "+perimeter);
      sc.close();
   }
}
```

OUTPUT:

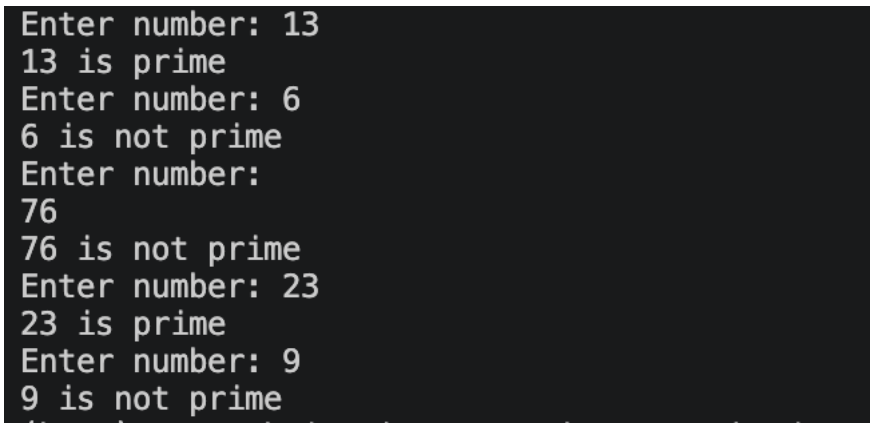


Q4.You are required to get five integer numbers from the keyboard and check whether these numbers are primes or not.
A4. CODE:

```
import java.util.Scanner;

public class P4{
   public static void main(String[] args){
      Scanner sc=new Scanner(System.in);
      for(int i=0;i<5;i++){
         System.out.print("Enter number: ");
         int n=sc.nextInt();
         boolean prime=true;
         if(n<2) prime=false;
         else{
            for(int j=2;j*j<=n;j++){
               if(n%j==0){
                  prime=false;
                  break;
               }
            }
         }
         if(prime) System.out.println(n+" is prime");
         else System.out.println(n+" is not prime");
      }
      sc.close();
   }
}
```

OUTPUT:



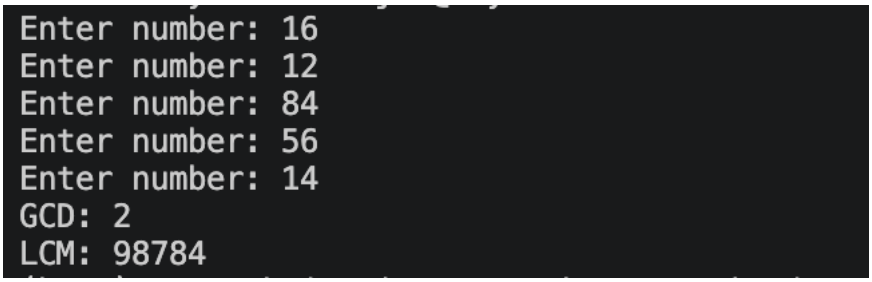
Q5. Repeat problem 4 for computing greatest common divisor (GCD) and least common multiple (LCM) for five integers entered from keyboard. (You should not use the concept of array)
A5. CODE:

```
import java.util.Scanner;

public class P5{
   public static void main(String[] args){
      Scanner sc=new Scanner(System.in);
      System.out.print("Enter number: ");
      int n=sc.nextInt();
      int gcd=n;
      int lcm=n;
      for(int i=1;i<5;i++){
         System.out.print("Enter number: ");
         n=sc.nextInt();
         int a=gcd;
         int b=n;
         while(b!=0){
            int temp=b;
            b=a%b;
            a=temp;
         }
         gcd=a;
         lcm=lcm/gcd*n;
      }
      System.out.println("GCD: "+gcd);
      System.out.println("LCM: "+lcm);

      sc.close();
   }
}
```

OUTPUT:



Q6. You are required to convert your marks evaluated out of 100 to the corresponding grades used in MNNIT Allahabad with the use of if-else control statement.

A6. CODE:

```
import java.util.Scanner;

public class P6{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        System.out.print("Enter marks: ");
        double marks=sc.nextDouble();

        if(marks>=85)
            System.out.println("Grade: A+");
        else if(marks>=75)
            System.out.println("Grade: A");
        else if(marks>=65)
            System.out.println("Grade: B+");
        else if(marks>=55)
            System.out.println("Grade: B");
        else if(marks>=45)
            System.out.println("Grade: C");
        else if(marks>=35)
            System.out.println("Grade: D");
        else
            System.out.println("Grade: F");

        sc.close();
    }
}
```

OUTPUT:

```
Enter marks: 54
Grade: C
```

Q7. Repeat the problem 6 with the use of switch control statement.

A7. CODE:

```
import java.util.Scanner;

public class P7{
   public static void main(String[] args){
      Scanner sc=new Scanner(System.in);

      System.out.print("Enter marks: ");
      int marks=sc.nextInt();

      int grade;

      if(marks>=85) grade=1;
      else if(marks>=75) grade=2;
      else if(marks>=65) grade=3;
      else if(marks>=55) grade=4;
      else if(marks>=45) grade=5;
      else if(marks>=35) grade=6;
      else grade=7;

      switch(grade){
         case 1:
            System.out.println("Grade: A+");
            break;
         case 2:
            System.out.println("Grade: A");
            break;
         case 3:
            System.out.println("Grade: B+");
            break;
         case 4:
            System.out.println("Grade: B");
            break;
         case 5:
            System.out.println("Grade: C");
            break;
         case 6:
            System.out.println("Grade: D");
            break;
         case 7:
            System.out.println("Grade: F");
            break;
      }

      sc.close();
   }
}
```

OUTPUT:



Q8. You are required to model simple digital lock with integer digit combination of three security keys. The combination is set into unlock when it is created security keys. To open the lock user must give correct combination given in term of single digit at a time. The lock explicitly remembers last three digits entered.

A8. CODE:

```
import java.util.Scanner;

public class P8{
   public static void main(String[] args){
      Scanner sc=new Scanner(System.in);

      System.out.println("Set 3 security keys:");
      int k1=sc.nextInt();
      int k2=sc.nextInt();
      int k3=sc.nextInt();
      int a=-1,b=-1,c=-1;
      System.out.println("Enter digits one at a time:");
      while(true){
         int digit=sc.nextInt();
         a=b;
         b=c;
         c=digit;
         if(a==k1 && b==k2 && c==k3){
            System.out.println("Lock opened");
            break;
         }
      }

      sc.close();
   }
}
```

OUTPUT:

```
Set 3 security keys:
5 4 6
Enter digits one at a time:
7 6 5 4 6
Lock opened
```

```
Set 3 security keys:
4 7 4
Enter digits one at a time:
3 4 7 4 7 6
Lock opened
```

Q9. You are required to model simple digital lock with integer digit combination of three security keys. The combination is set into unlock when it is created security keys. To open the lock user must give correct combination given in term of single digit at a time. The lock instead of remembering last three digits entered to make the lock open. The lock remembers if last digit entered was first security key and if last two digits were first two security keys.
A9. CODE:

```
import java.util.Scanner;

public class P9{
   public static void main(String[] args){
      Scanner sc=new Scanner(System.in);

      System.out.println("Set 3 security keys:");
      int k1=sc.nextInt();
      int k2=sc.nextInt();
      int k3=sc.nextInt();

      int state=0;

      System.out.println("Enter digits one at a time:");

      while(true){
         int digit=sc.nextInt();

         if(state==0){
            if(digit==k1){state=1;}
         }
         else if(state==1){
            if(digit==k2){state=2;}
            else if(digit!=k1){state=0;}
         }
         else if(state==2){
            if(digit==k3){System.out.println("Lock opened");break;}
            else if(digit==k1){state=1;}
            else{state=0;}
         }
      }
      sc.close();
   }
}
```

OUTPUT:

